

Ref. No: CUIET/CSE/ACAD/2026/97

Date: 02/04/2026

NOTICE
Attention:- B.E. CSE Batch 2023

Test Type: ST **Test No:** 1 **Total Marks:** 60 **Duration:** 60 minutes

Name of Course Coordinator: Dr. Suhasini

Course Code: 23CS020

Course Title: Advanced System Design

Day: Thursday

Date: 09/04/2026

Time: 9:00AM – 04:00PM

Mode: Online (On-Campus)

Semester: B.E.-CSE 6th Sem

Platform: Testpad

Groups:- G1-G12, G14,G16, G17-27,G34,G36

Specialization: Full Stack Engineering, University Coding Academy,Japanese

Syllabus for Test: (As per CHO): Foundation of Large Scale Systems Design(Recap): System design principles, Functional versus non-functional requirements, Latency vs throughput, Design Approaches and Architectural Patterns, **Fundamentals of Distributed Systems:** Definition and characteristics of distributed systems, Advantages and challenges, Types of distributed systems, Client-server and peer-to-peer models, Distributed System Failures: Failure models & fault tolerance, **Advanced Distributed Systems Concepts:** Process coordination and Synchronization , Mutual exclusion, Deadlock Handling, Parallel Computing, Distributed Computing, Parallel Computing versus Distributed Computing, CAP theorem, PACELC theorem, Revision of Consistency Models, Consensus algorithms (Raft, Paxos – internal working), Leader election and failure detection

Format of Assessment:

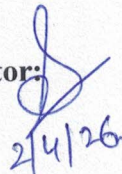
1 Mark MCQ: 30

2 Mark MCQ: 15

Total MCQ: 45

Total Marks: 60

Signature of Course Coordinator:



24/26